CLASS X : CHAPTER - 12
AREAS RELATED TO CIRCLES

IMPORTANT FORMULAS & CONCEPTS

Perimeter and Area of a Circle

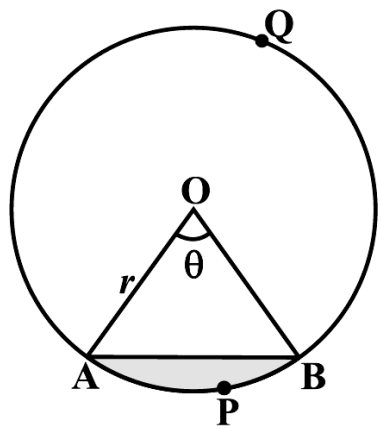
Perimeter/circumference of a circle $= \pi \times$ diameter
$= \pi \times 2r$ (where r is the radius of the circle)
$= 2\pi r$

Area of a circle $= \pi r^2$, where $\pi = \frac{22}{7}$

Areas of Sector and Segment of a Circle

Area of the sector of angle $\theta = \frac{\theta}{360^0} \times \pi r^2$, where r is the radius of the circle and θ the angle of the sector in degrees

length of an arc of a sector of angle $\theta = \frac{\theta}{360^0} \times 2\pi r$, where r is the radius of the circle and θ the angle of the sector in degrees



Area of the segment APB = Area of the sector OAPB – Area of Δ OAB

$= \frac{\theta}{360^0} \times \pi r^2 -$ area of Δ OAB

- Area of the major sector OAQB = πr^2 – Area of the minor sector OAPB
- Area of major segment AQB = πr^2 – Area of the minor segment APB
- Area of segment of a circle = Area of the corresponding sector – Area of the corresponding triangle

MCQ WORKSHEET-I
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AREAS RELATED TO CIRCLES

1. The area of a circle is 49π cm^2. Its circumference is
(a) 7π cm (b) 14π cm (c) 21π cm (d) 28π cm

2. The perimeter of circular field is 242cm. The area of the field is
(a) 9317 cm^2 (b) 18634 cm^2 (c) 4658.5 cm^2 (d) none of these

3. The area of a circle is 38.5 cm^2. Its circumference is
(a) 62 cm (b) 12.1 cm (c) 11 cm (d) 22 cm

4. The difference between the circumference and radius of a circle is 37 cm. The area of the circle is
(a) 111 cm^2 (b) 184 cm^2 (c) 154 cm^2 (d) 259 cm^2

5. The circumference of two circles are in the ratio 2 : 3. The ratio of their areas is
(a) 2 : 3 (b) 4 : 9 (c) 9 : 4 (d) none of these

6. On increasing the diameter of circle by 40%, its area will be increased by
(a) 40% (b) 80% (c) 96% (d) none of these

7. On decreasing the radius of the circle by 30%, its area is decreased by
(a) 30% (b) 60% (c) 45% (d) none of these

8. The area of the square is the same as the area of the circle. Their perimeter re in the ratio
(a) 1 : 1 (b) π : 2 (c) 2 : π (d) none of these

9. The areas of the two circle are in the ratio 4 : 9. The ratio of their circumference is
(a) 2 : 3 (b) 4 : 9 (c) 9 : 4 (d) 4 : 9

10. In making 1000 revolutions, a wheel covers 88 km. The diameter of the wheel is
(a) 14 m (b) 24 m (c) 28 m (d) 40 m

11. The diameter of a wheel is 40 cm. How many revolutions will it make an covering 176 m?
(a) 140 (b) 150 (c) 160 (d) 166

12. The radius of wheel is 0.25 m. How many revolutions will it make in covering 11 km?
(a) 2800 (b) 4000 (c) 5500 (d) 7000

13. Find the circumference of a circle of diameter 21 cm.
(a) 62 cm (b) 64 cm (c) 66 cm (d) 68 cm

14. Find the area of a circle whose circumference is 52.8 cm.
(a) 221.76 cm^2 (b) 220.76 cm^2 (c) 200.76 cm^2 (d) none of these.

15. A steel wire when bent in the form of a square, encloses an area of 121 sq. cm. The same wire is bent in the form of a circle. Find the area of the circle.
(a) 111 cm^2 (b) 184 cm^2 (c) 154 cm^2 (d) 259 cm^2

MCQ WORKSHEET-II
CLASS X: CHAPTER – 12
AREAS RELATED TO CIRCLES

1. A wire is looped in the form of a circle of radius 28 cm. It is rebent into a square form. Determine the length of the side of the square.
(a) 42 cm (b) 44 cm (c) 46 cm (d) 48 cm

2. A circular part, 42 m in diameter has a path 3.5 m wide running round it on the outside. Find the cost of gravelling the path at Rs. 4 per m^2.
(a) Rs. 2800 (b) Rs. 2020 (c) Rs. 2002 (d) none of these

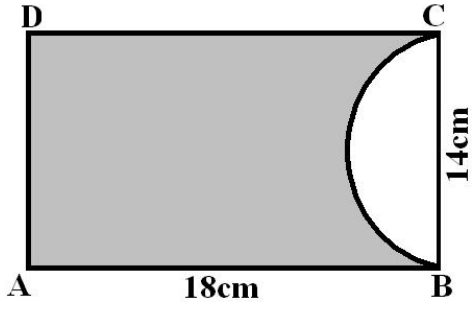
3. A road which is 7m wide surrounds a circular park whose circumference is 352 m. Find the area of the road.
(a) 2618 m^2 (b) 2518 m^2 (c) 1618 m^2 (d) none of these

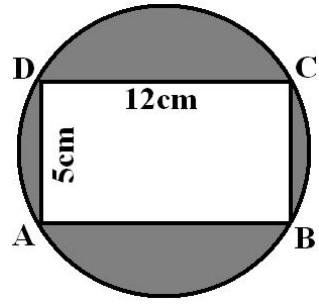
4. If the perimeter of a semicircular protractor is 36 cm, find the diameter.
(a) 14 cm (b) 16 cm (c) 18 cm (d) 12 cm

5. A bicycle wheel makes 5000 revolutions in moving 11 km. Find the diameter of the wheel.
(a) 60 cm (b) 70 cm (c) 66 cm (d) 68 cm

6. The diameter of the wheels of a bus is 140 cm. How many revolutions per minute must a wheel make in order to move a t a speed of 66km/hr?
(a) 240 (b) 250 (c) 260 (d) 270

7. A paper is in the form of a rectangle ABCD in which AB = 18cm and BC = 14cm. A semicircular portion with BC as diameter is cut off. Find the area of the remaining paper (see in below figure).
(a) 175 cm^2 (b) 165 cm^2 (c) 145 cm^2 (d) none of these





8. Find the area of the shaded region in the above sided figure. Take $\pi = 3.14$
(a) 75 cm^2 (b) 72 cm^2 (c) 70 cm^2 (d) none of these

9. A square ABCD is inscribed in a circle of radius 'r'. Find the area of the square in sq. units.
(a) $3r^2$ (b) $2r^2$ (c) $4r^2$ (d) none of these

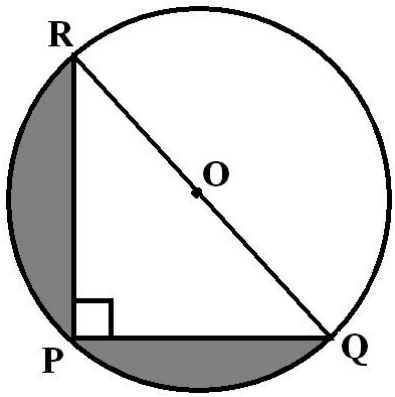
10. Find the area of a right-angled triangle, if the radius of its circumcircle is 2.5 cm and the altitude drawn to the hypotenuse is 2 cm long.
(a) 5 cm^2 (b) 6 cm^2 (c) 7 cm^2 (d) none of these

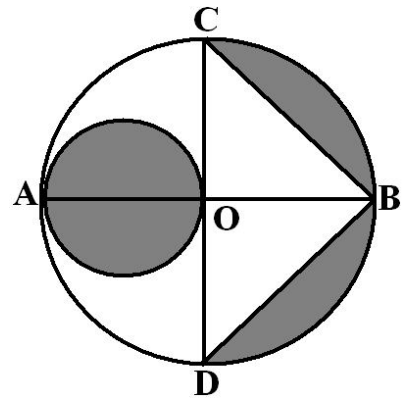
MCQ WORKSHEET-III
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AREAS RELATED TO CIRCLES

1. The perimeter of a sector of a circle of radius 5.6 cm is 27.2 cm. Find the area of the sector.
(a) 44 cm^2 (b) 44.6 cm^2 (c) 44.8 cm^2 (d) none of these

2. The minute hand of a clock is 12 cm long. Find the area of the face of the clock described by the minute hand in 35 minutes.
(a) 265 cm^2 (b) 266 cm^2 (c) 264 cm^2 (d) none of these

3. Find the area of the shaded region in the given figure, if Pr = 24 cm, PQ = 7 cm and O is the centre of the circle.
(a) 164.54 cm^2 (b) 161.54 cm^2 (c) 162.54 cm^2 (d) none of these





4. In the above-sided figure, AB is a diameter of a circle with centre O and OA = 7 cm. Find the area of the shaded region.
(a) 64.5 cm^2 (b) 61.5 cm^2 (c) 66.5 cm^2 (d) none of these

5. A racetrack is in the form of a ring whose inner circumference is 352 m and outer circumference is 396 m. Find the width of the track.
(a) 4 m (b) 6 m (c) 8 m (d) 7 m

6. The difference between the circumference and the radius of a circle is 37 cm. Find the area of the circle.
(a) 111 cm^2 (b) 184 cm^2 (c) 154 cm^2 (d) 259 cm^2

7. The circumference of a circle exceeds its diameter by 16.8 cm. Find the circumference of the circle.
(a) 24.64 cm (b) 26.64 cm (c) 28.64 cm (d) 22 cm

8. A copper wire when bent in the form of square encloses an area of 484 cm^2. The same wire is now bent in the form of a circle. Find the area of the circle.
(a) 116 cm^2 (b) 166 cm^2 (c) 616 cm^2 (d) none of these

9. Find the area of the sector of a circle of radius 14 cm with central angle 45^0.
(a) 76 cm^2 (b) 77 cm^2 (c) 66 cm^2 (d) none of these

10. A sector is cut from a circle of radius 21 cm. The angle of the sector is 150^0. Find the length of the arc.
(a) 56 cm (b) 57 cm (c) 55 cm (d) 58 cm

MCQ WORKSHEET-IV
CLASS X: CHAPTER – 12
AREAS RELATED TO CIRCLES

1. The area of the sector of a circle of radius r and central angle θ, is

A. ½ l.r B. $2\pi r^2\theta/720$ C. $2\pi r\theta/360$ D. $\pi r\theta/360$

2. An arc of a circle is of length 5π cm and the sector it bounds has an area of 20π cm^2. The radius of circle is

A. 1 cm B. 5 cm C. 8 cm D. 10 cm

3. A sector is cut from a circle of circle of radius 21 cm. The angle of sector is 150°. The area of sector is

A. 577.5 cm^2 B. 288.2 cm^2 C. 152 cm^2 D. 155 m^2

4. A chord AB of a circle of radius 10 cm makes a right angle at the centre of the circle. The area of major segment is

A. 210 cm^2 B. 235.7 cm^2 C. 185.5 cm^2 D. 258.1 cm^2

5. A horse is tied to a pole with 56 m long string. The area of the field where the horse can graze is

A. 2560 m^2 B. 2464 m^2 C. 9856 m^2 D. 25600 m^2

6. The circumferences of two circles are in the ratio 2:3. The ratio of their areas is

A. 4:9 B. 2:3 C. 7:9 D. 4:10

7. Area enclosed between two concentric circles is 770 cm^2. If the radius of outer circle is 21 cm, then the radius of inner circle is

A. 12 cm B. 13 cm C. 14 cm D. 15 cm

8. The perimeter of a semi-circular protector is 72 cm. Its diameter is

A. 28 cm B. 14 cm C. 36 cm D. 24 cm

9. The minute hand of a clock is 21 cm long. The area described by it on the face of clock in 5 minutes is

A. 115.5 cm^2 B. 112.5 cm^2 C. 211.5 cm^2 D. 123.5 cm^2

10. The area of a circle circumscribing a square of area 64 cm^2 is

A. 50.28 cm^2 B. 25.5 cm^2 C. 100.57 cm^2 D. 75.48 cm^2

11. A pendulum swings through an angle of 30^0 and describes an arc 8.8 cm in length. Find the length of the pendulum.

(a) 16 cm (b) 16.5 cm (c) 16.8 cm (d) 17 cm

12. The minute hand of a clock is 15 cm long. Calculate the area swept by it in 20 minutes. Take $\pi=3.14$

(a) 116 cm^2 (b) 166 cm^2 (c) 616 cm^2 (d) none of these

13. A sector of 56^0, cut out from a circle, contains 17.6 cm^2. Find the radius of the circle.

(a) 6 cm (b) 7 cm (c) 5 cm (d) 8 cm

14. A chord 10 cm long is drawn in a circle whose radius is $5\sqrt{2}$ cm. Find the areas of minor segment. Take $\pi=3.14$

(a) 16 cm^2 (b) 14.5 cm^2 (c) 14.25 cm^2 (d) none of these

15. The circumference of a circle is 88 cm. Find the area of the sector whose central angle is 72^0.

(a) 123 cm^2 (b) 123.5 cm^2 (c) 123.4 cm^2 (d) none of these

PRACTICE QUESTIONS
CLASS X: CHAPTER – 12
AREAS RELATED TO CIRCLES

1. If the perimeter of a semicircular protractor is 36 cm, find its diameter.

2. A bicycle wheel makes 5000 revolutions in moving 11 km. Find the diameter of the wheel.

3. The diameter of the wheels of a bus is 140 cm. How many revolutions per minute must a wheel make in order to move at a speed of 66 km per hour?

4. Two circles touch externally. The sum of their areas is 130π sq. cm and the distance between their centres is 14 cm. Find the radii of the circles.

5. Two circles touch internally. The sum of their areas is 116π sq. cm and the distance between their centres is 6 cm. Find the radii of the circles.

6. A paper is in the form of a rectangle ABCD in which AB = 18 cm and BC = 14 cm. A semicircular portion with BC as diameter is cut off. Find the area of the remaining paper.

7. A square ABCD is inscribed in a circle of radius r. Find the area of the square.

8. Find the area of a right-angled triangle, if the radius of its circumcircle is 2.5cm and the altitude drawn to the hypotenuse is 2cm long.

9. A steel wire, bent in the form of a square, encloses an area of 121 sq. cm. The same wire is bent in the form of a circle. Find the area of the circle.

10. A wire is looped in the form of a circle of radius 28 cm. It is rebent into a square form. Determine the length of the side of the square.

11. A circular park, 42 m diameter, has a path 3.5 m wide running round it on the outside. Find the cost of gravelling the path at Rs. 4 per m^2.

12. A road, which is 7m wide, surrounds a circular park whose circumference is 352m. Find the area of the road.

13. A racetrack is in the form of a ring whose inner and outer circumference are 437 m and 503 m respectively. Find the width of the track and also it area.

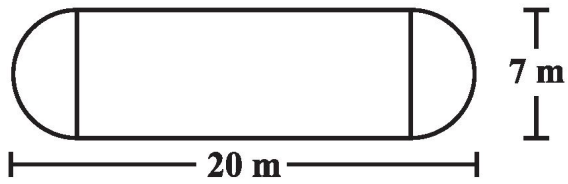
14. From a circular sheet of radius 4 cm, a circle of radius 3 cm is removed. Find the area of the remaining sheet. (Take $\pi = 3.14$)

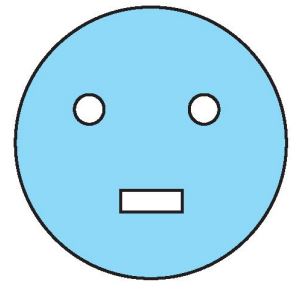
15. Saima wants to put a lace on the edge of a circular table cover of diameter 1.5 m. Find the length of the lace required and also find its cost if one meter of the lace costs Rs 15. (Take $\pi = 3.14$)

16. A circle of radius 2 cm is cut out from a square piece of an aluminium sheet of side 6 cm. What is the area of the left over aluminium sheet? (Take $\pi = 3.14$)

17. The circumference of a circle is 31.4 cm. Find the radius and the area of the circle? (Take $\pi = 3.14$)

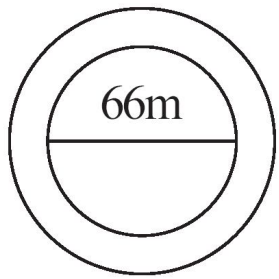
18. The shape of a garden is rectangular in the middle and semi circular at the ends as shown in the diagram. Find the area and the perimeter of this garden

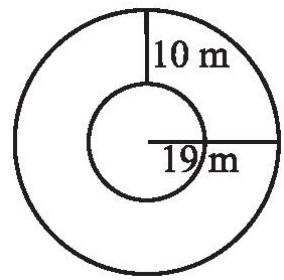




19. From a circular card sheet of radius 14 cm, two circles of radius 3.5 cm and a rectangle of length 3 cm and breadth 1cm are removed. (as shown in the right sided adjoining figure). Find the area of the remaining sheet.

20. A circular flower bed is surrounded by a path 4 m wide. The diameter of the flower bed is 66 m. What is the area of this path? (π= 3.14)





21. Find the circumference of the inner and the outer circles, shown in the right sided adjoining figure? (Take π= 3.14)

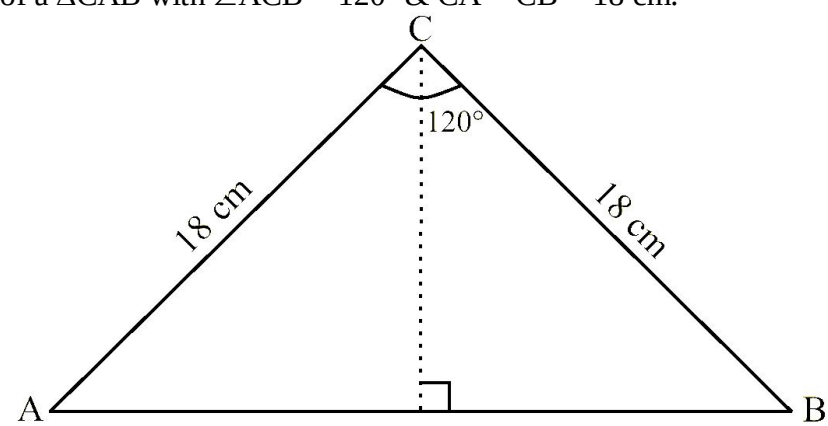
22. Shazli took a wire of length 44 cm and bent it into the shape of a circle. Find the radius of that circle. Also find its area. If the same wire is bent into the shape of a square, what will be the length of each of its sides? Which figure encloses more area, the circle or the square?

23. A circular flower garden has an area of 314 m^2. A sprinkler at the centre of the garden can cover an area that has a radius of 12 m. Will the sprinkler water the entire garden? (Take π = 3.14)

24. How many times a wheel of radius 28 cm must rotate to go 352 m? (Take $\pi = \frac{22}{7}$)

25. Three horses are tethered with 7 m long ropes at the three corners of a triangular field having sides 20m, 34 m and 42 m. Find the area of the plot which can be grazed by the horses. Also, find the area of the plot, which remains ungrazed.

26. Find the area of a ΔCAB with $\angle ACB = 120^0$ & CA = CB = 18 cm.



27. Find the area of sector of angle 120^0 and radius 18 cm.

28. Find the area of the segment AOB of angle 120^0 and radius 18 cm.

29. The minute hand of a circular clock is 15 cm long. Find the area of the face of the clock and how far does the tip of the minute hand move in 35 minutes? (Take π= 3.14)

30. Find the cost of polishing a circular table-top of diameter 1.6 m, if the rate of polishing is Rs 15/m^2. (Take π = 3.14)

31. A chord of a circle of radius 14 cm makes a right angle at the centre. Find the areas of the minor and the major segments of the circle.

32. A square tank has area of 1600 m^2. There are four semicircular plots around it. Find the cost of turfing the plots at Rs. 1.25 per m^2. Take π = 3.14.

33. A lawn is rectangular in the middle and it has semicircular portions along the shorter sides of the rectangle. The rectangular portion measures 50m by 35m. Find the area of the lawn.

34. A rope by which a cow is tethered is increased from 16 m to 23 m. How much additional ground does it have now to graze?

35. The perimeter of a certain sector of a circle of radius 6.5 cm is 31 cm. Find the area of the sector.

36. The area of the sector of a circle of radius 10.5 cm is 69.3 cm^2. Find the central angle of the sector.

37. A sector of 56^0 cut out from a circle, contains 17.6 cm^2. Find the radius of the circle.

38. The short and long hands of a clock are 4 cm and 6 cm long respectively. Find the sum of distances travelled be their tips in 2 days. Take π = 3.14.

39. Find the lengths of the arcs cut off from a circle of radius 12 cm by a chord 12 cm long. Also find the area of the minor segment. Take $\sqrt{3}$ =1.73 and π = 3.14.

40. The perimeter of a sector of a circle of radius 5.6 cm is 27.2 cm. Find the area of the sector.

41. A chord of a circle of radius 10 cm subtends a right angle at the centre. Find the following: (i) Area of minor sector (ii) Area of major sector (iii) Area of major segment (iv) Area of minor segment. (Use π = 3.14)

42. In a circle of radius 10.5 cm, the minor arc is one-fifth of the major arc. Find the area of the sector corresponding to the major arc.

43. It is proposed to add two circular ends, to a square lawn whose side measures 58 cm, the centre of each circle being the point of intersection of the diagonals of the square. Find the area of the whole lawn.

44. It is proposed to add two circular ends, to a square lawn whose side measures 50 m, the centre of each circle being the point of intersection of the diagonals of the square. Find the area of the whole lawn. Take π = 3.14

45. In an equilateral triangle of side 12 cm, a circle is inscribed touching its sides. Find the area of the portion of the triangle not included in the circle. Take $\sqrt{3}$ =1.73 and π = 3.14.

46. In a circle of radius 21 cm, an arc subtends an angle of 60^0 at the centre. Find (i) length of the arc (ii) area of sector formed by the arc (iii) area of segment formed by the corresponding chord of the arc.

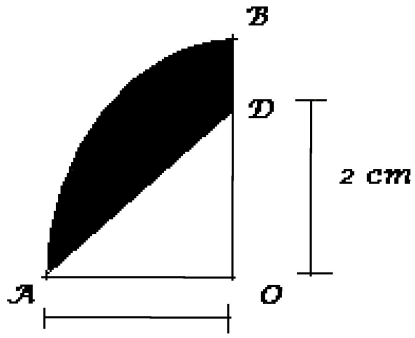
47. If three circles of radius r each, are drawn such that each touches the other two, the find the area included between them. Take $\pi = 3.14$ and $\sqrt{3} = 1.73$.

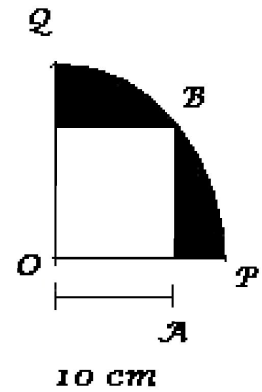
48. If four circles of radius r each, are drawn such that each touches the other two, the find the area included between them. Take $\pi = 3.14$.

49. The length of an arc subtending an angle of 72^0 at the centre is 44 cm. Find the area of the circle.

50. A park is in the form of rectangle 120 m by 100 m. At the centre of the park, there is a circular lawn. The area of the park excluding the lawn is 11384 sq. m. Find the radius of the circular lawn.

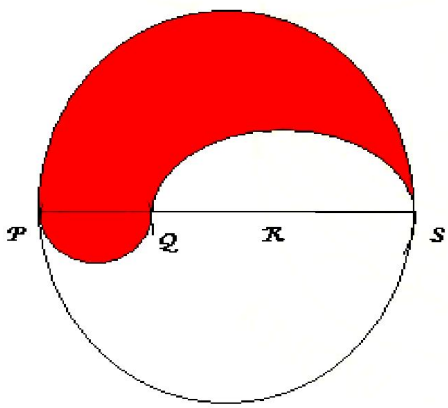
51. Find the area of shaded portion in the below figure

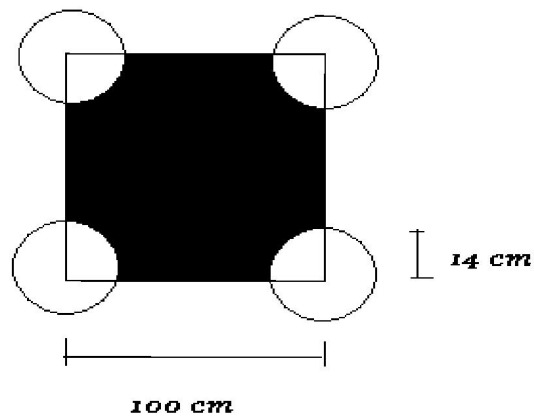




52. Find the area of shaded portion in the above right-sided figure

53. Find the area of shaded portion in the below figure





54. Find the area of shaded portion in the above right-sided figure

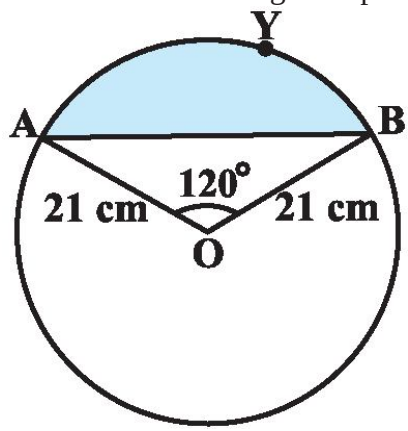
55. An athletic track, 14 m wide, consists of two straight sections 120 m long joining semicircular ends whose inner radius is 35 m. Calculate the area of the track.

56. The cost of fencing a circular field at the rate of Rs 24 per metre is Rs 5280. The field is to be ploughed at the rate of Rs 0.50 per m2. Find the cost of ploughing the field.

57. The radii of two circles are 8 cm and 6 cm respectively. Find the radius of the circle having area equal to the sum of the areas of the two circles.

58. The radii of two circles are 19 cm and 9 cm respectively. Find the radius of the circle which has circumference equal to the sum of the circumferences of the two circles.

59. The wheels of a car are of diameter 80 cm each. How many complete revolutions does each wheel make in 10 minutes when the car is travelling at a speed of 66 km per hour?



60. Find the area of the segment AYB shown in Fig., if radius of the circle is 21 cm and $\angle AOB = 120°$. (Use $\pi = 22/7$).

61. Find the area of the sector of a circle with radius 4 cm and of angle 30°. Also, find the area of the corresponding major sector (Use $\pi = 3.14$).

62. Find the area of a sector of a circle with radius 6 cm if angle of the sector is 60°.

63. Find the area of a quadrant of a circle whose circumference is 22 cm.

64. The length of the minute hand of a clock is 14 cm. Find the area swept by the minute hand in 5 minutes.

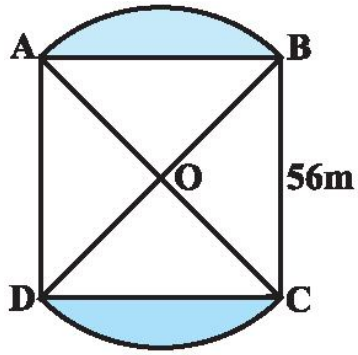
65. A chord of a circle of radius 10 cm subtends a right angle at the centre. Find the area of the corresponding : (i) minor segment (ii) major sector. (Use $\pi = 3.14$)

66. In a circle of radius 21 cm, an arc subtends an angle of 60° at the centre. Find: (i) the length of the arc (ii) area of the sector formed by the arc (iii) area of the segment formed by the corresponding chord

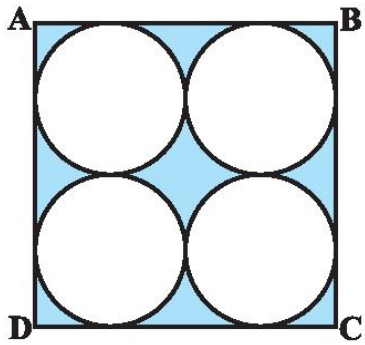
67. A chord of a circle of radius 15 cm subtends an angle of 60° at the centre. Find the areas of the corresponding minor and major segments of the circle. (Use $\pi = 3.14$ and $\sqrt{3} = 1.73$)

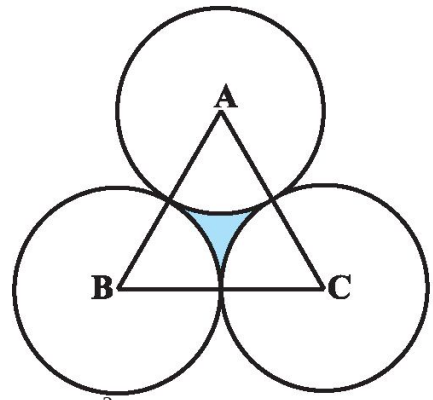
68. A chord of a circle of radius 12 cm subtends an angle of 120° at the centre. Find the area of the corresponding segment of the circle. (Use $\pi = 3.14$ and $\sqrt{3} = 1.73$)

69. In Fig, two circular flower beds have been shown on two sides of a square lawn ABCD of side 56 m. If the centre of each circular flower bed is the point of intersection O of the diagonals of the square lawn, find the sum of the areas of the lawn and the flower beds.



70. Find the area of the shaded region in the below Fig., where ABCD is a square of side 14 cm.



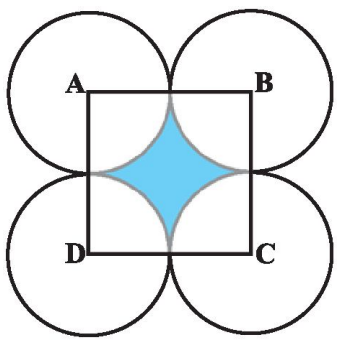


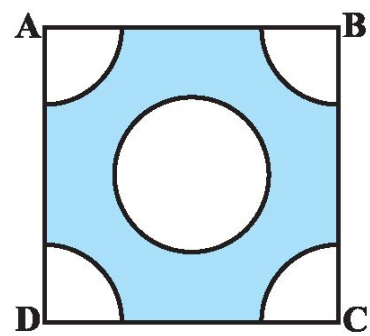
71. The area of an equilateral triangle ABC is 17320.5 cm^2. With each vertex of the triangle as centre, a circle is drawn with radius equal to half the length of the side of the triangle. Find the area of the shaded region. . (Use $\pi = 3.14$ and $\sqrt{3} = 1.73205$)

72. An umbrella has 8 ribs which are equally spaced. Assuming umbrella to be a flat circle of radius 45 cm, find the area between the two consecutive ribs of the umbrella.

73. A horse is tied to a peg at one corner of a square shaped grass field of side 15 m by means of a 5 m long rope. Find (i) the area of that part of the field in which the horse can graze. (ii) the increase in the grazing area if the rope were 10 m long instead of 5 m. . (Use $\pi = 3.14$)

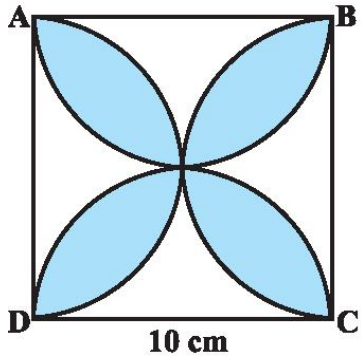
74. In Fig., ABCD is a square of side 14 cm. With centres A, B, C and D, four circles are drawn such that each circle touch externally two of the remaining three circles. Find the area of the shaded region.

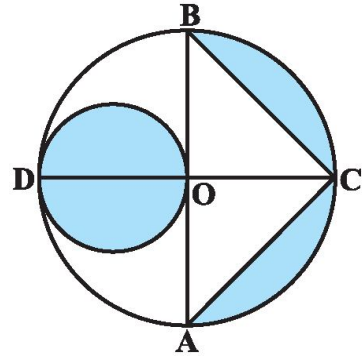




75. From each corner of a square of side 4 cm a quadrant of a circle of radius 1 cm is cut and also a circle of diameter 2 cm is cut as shown in above sided Fig. Find the area of the remaining portion of the square.

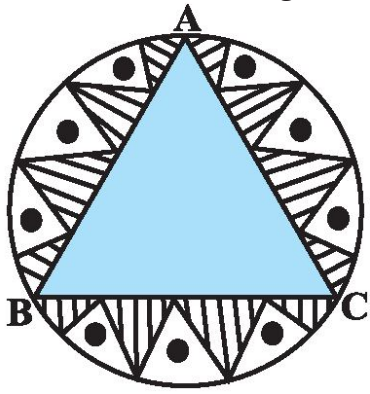
76. Find the area of the shaded design in the below Fig., where ABCD is a square of side 10 cm and semicircles are drawn with each side of the square as diameter. (Use $\pi = 3.14$)

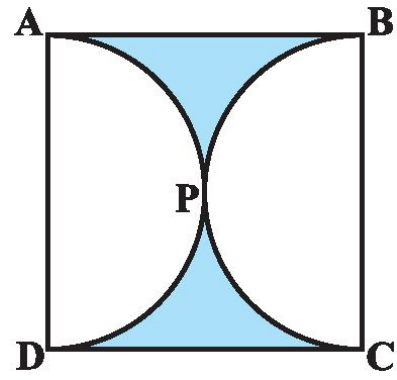




77. In the above sided Fig., AB and CD are two diameters of a circle (with centre O) perpendicular to each other and OD is the diameter of the smaller circle. If OA = 7 cm, find the area of the shaded region.

78. In a circular table cover of radius 32 cm, a design is formed leaving an equilateral triangle ABC in the middle as shown in Fig. Find the area of the design (shaded region).

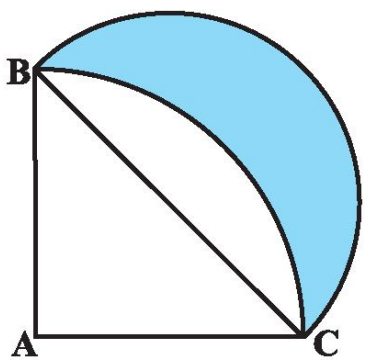


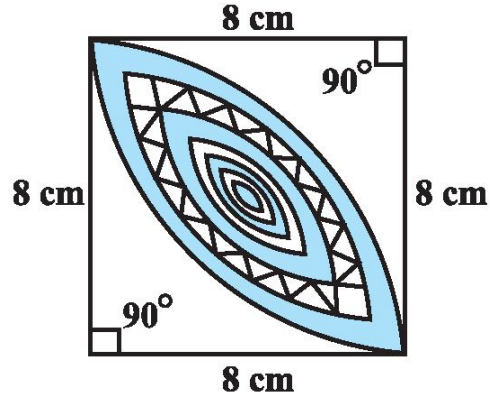


79. Find the area of the shaded region in above sided Fig., if ABCD is a square of side 14 cm and APD and BPC are semicircles.

80. To warn ships for underwater rocks, a lighthouse spreads a red coloured light over a sector of angle 80° to a distance of 16.5 km. Find the area of the sea over which the ships are warned. . (Use $\pi = 3.14$)

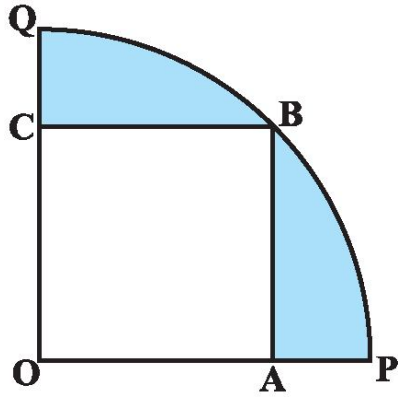
81. In Fig., ABC is a quadrant of a circle of radius 14 cm and a semicircle is drawn with BC as diameter. Find the area of the shaded region.

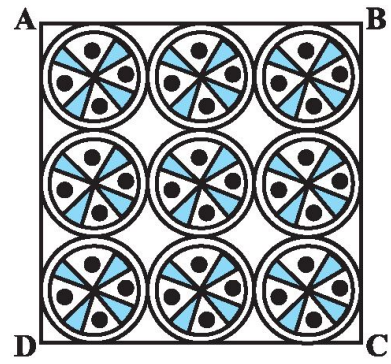




82. Calculate the area of the designed region in the above sided Fig. common between the two quadrants of circles of radius 8 cm each.

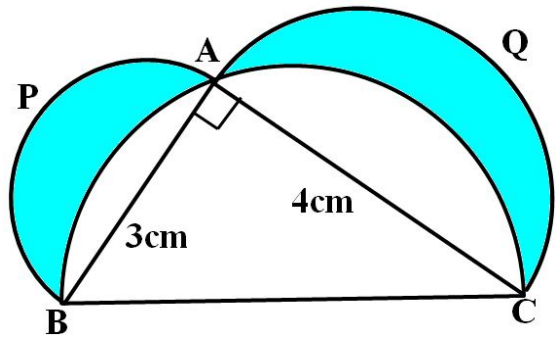
83. In Fig., a square OABC is inscribed in a quadrant OPBQ. If OA = 20 cm, find the area of the shaded region. (Use $\pi = 3.14$)



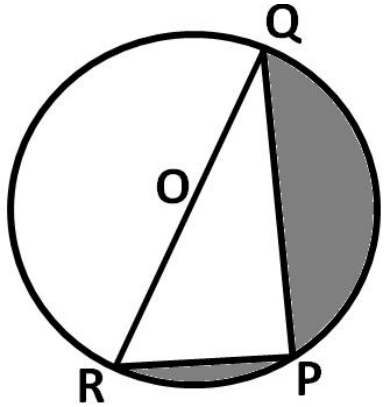


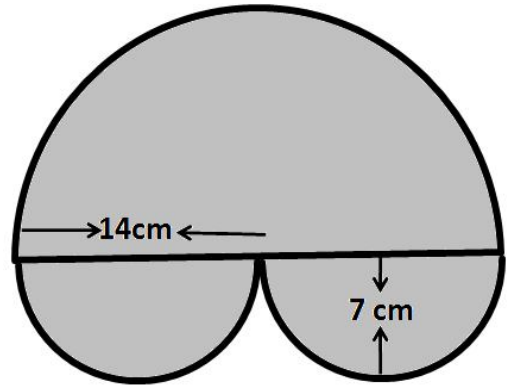
84. On a square handkerchief, nine circular designs each of radius 7 cm are made (see in the above sided Fig.). Find the area of the remaining portion of the handkerchief.

85. In the given figure, ΔABC is right angled at A. Semicircles are drawn on AB, AC and BC as diameters. It is given that AB = 3cm and AC = 4cm. Find the area of the shaded region.



86. Find the area of the shaded region in the below figure, if PQ = 24 cm, PR = 7 cm and O is the centre of the circle.





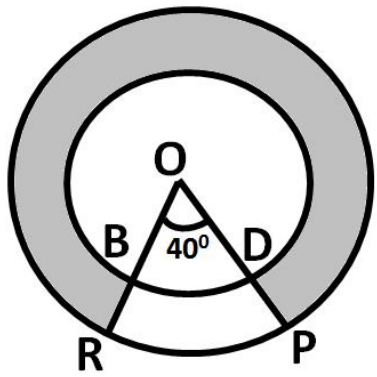
87. Find the areas of the shaded region in the above right sided figure.

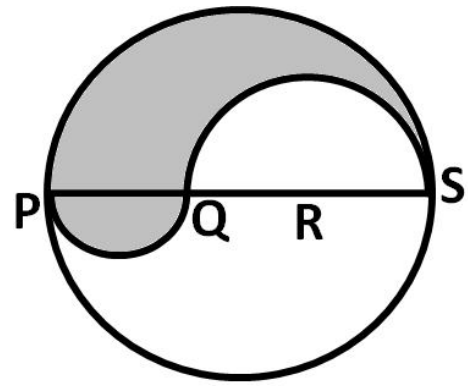
88. In an equilateral triangle of side 24 cm, a circle is inscribed touching its sides. Find the area of the remaining portion of the triangle. Take $\sqrt{3} = 1.732$

89. Find to the three places of decimals the radius of the circle whose area is the sum of the areas of two triangles whose sides are 35, 53, 66 and 33, 56, 65 measured in cms. (Take $\pi = \frac{22}{7}$)

90. A square park has each side of 100m. At each corner of the park, there is a slower bed in the form of a quadrant of radius 14 m. Find the area of the remaining part of the park. (Take $\pi = \frac{22}{7}$)

91. Find the area of the shaded region in below figure, where radii of the two concentric circles with centre O are 7 cm and 14 cm respectively and $\angle AOC = 40^0$.





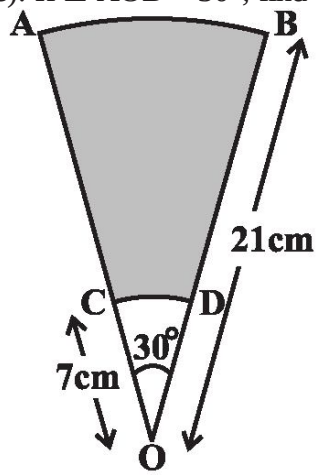
92. PQRS is a diameter of a circle of radius 6 cm. The lengths PQ, QR and RS are equal. Semicircles are drawn on PQ and QS as diameters as shown in above right sided figure. Find the perimeter and area of the shaded region.

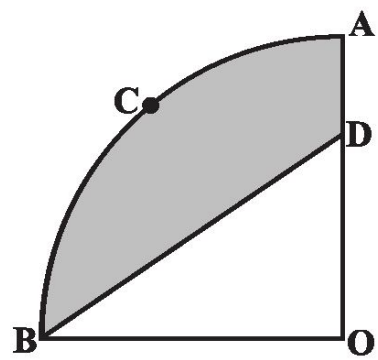
93. An athletic track 14 m wide consists of two straight sections 120 m long joining semicirculars ends whose inner radius is 35m. Calculate the area of the shaded region.



94. The above right-sided figure depicts a racing track whose left and right ends are semicircular. The distance between the two inner parallel line segments is 60 m and they are each 106 m long. If the track is 10 m wide, find : (i) the distance around the track along its inner edge (ii) the area of the track.

95. AB and CD are respectively arcs of two concentric circles of radii 21 cm and 7 cm and centre O (see below Figure). If $\angle$ AOB = 30°, find the area of the shaded region.



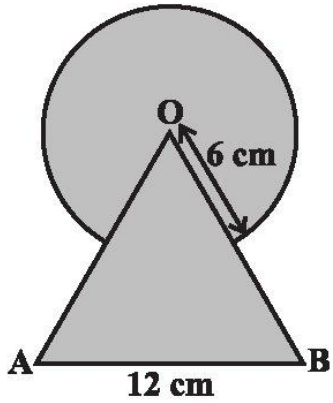


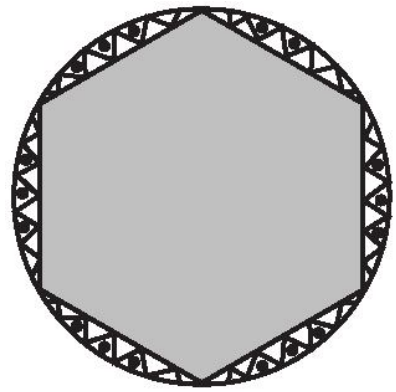
96. In the above right sided Figure, OACB is a quadrant of a circle with centre O and radius 3.5 cm. If OD = 2 cm, find the area of the (i) quadrant OACB, (ii) shaded region.

97. A path of 4m width runs round a semicircular grassy plot whose circumference is $163\frac{3}{7}$ m. Find

(i) the area of the path (ii) the cost of gravelling the path at the rate of Rs. 1.50 per sq. m (iii) the cost of turfing the plot at the rate of 45 paise per sq. m.

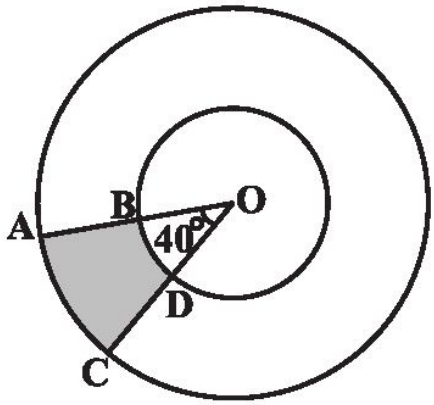
98. Find the area of the shaded region in below figure, where a circular arc of radius 6 cm has been drawn with vertex O of an equilateral triangle OAB of side 12 cm as centre.

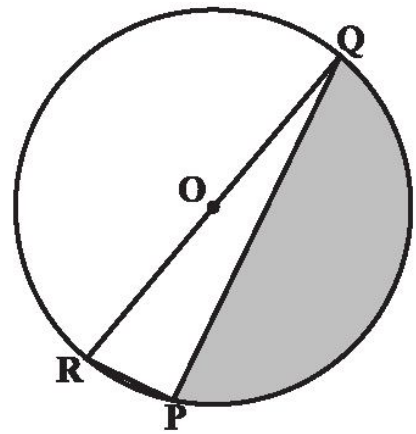




99. A round table cover has six equal designs as shown in the above right-sided figure. If the radius of the cover is 28 cm, find the cost of making the designs at the rate of Rs 0.35 per cm2. (Use $\sqrt{3}$ =1.7)

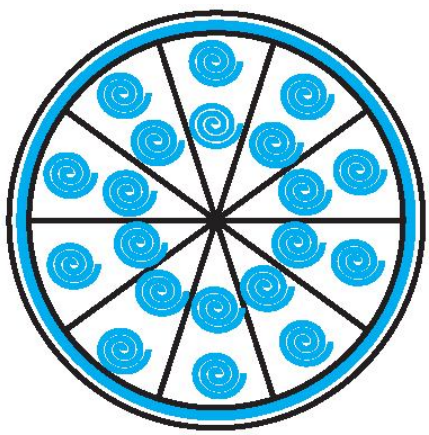
100. Find the area of the shaded region in the below figure, if radii of the two concentric circles with centre O are 7 cm and 14 cm respectively and ∠AOC = 40°.

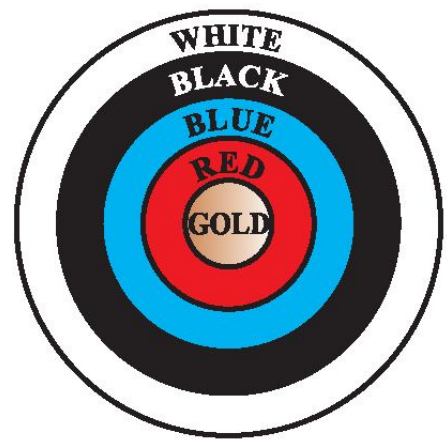




101. Find the area of the shaded region in the above right-sided figure, if PQ = 24 cm, PR = 7 cm and O is the centre of the circle.

102. A brooch is made with silver wire in the form of a circle with diameter 35 mm. The wire is also used in making 5 diameters which divide the circle into 10 equal sectors as shown in below figure. Find : (i) the total length of the silver wire required. (ii) the area of each sector of the brooch.

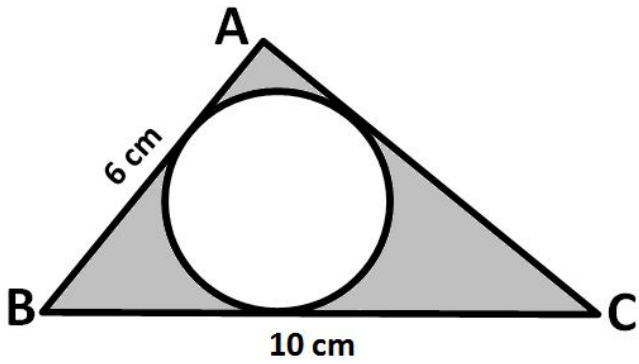


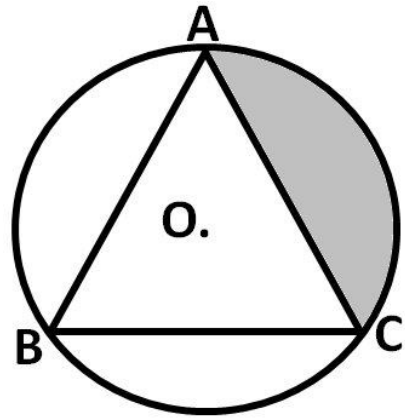


103. The area of a sector is one-twelfth that of the complete circle. Find the angle of the sectort.

104. Find the area of the circle in which a square of area 64 sq. cm is inscribed . (use π = 3.14)

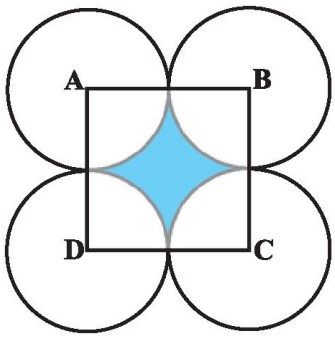
105. In the below figure, ABC is right angled triangle at A. Find the area of the shaded region, if AB = 6cm and BC = 10cm..

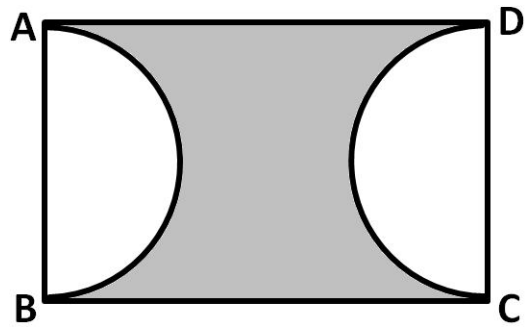




106. In the above right-sided figure, ABC is an equilateral triangle inscribed in a circle of radius 4 cm with centre O. Find the area of the shaded region.

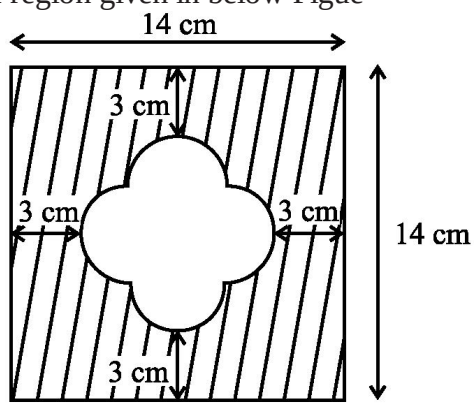
107. The diameter of a coin is 1 cm see the below figure. If four such coins be placed on a table so that the rim of each touches that of the other two, find the area of the shaded region. (use π = 3.1416)





108. In the above right-sided figure, ABCD is a rectangle, having AB = 14 cm and BC = 20 cm. Two sectors of 1800 have been cut off. Calculate (i) area of the shaded region (ii) length of the boundary of the shaded region.

109. Find the area of the shaded region given in below Figue



110. Find the number of revolutions made by a circular wheel of area 1.54 m2 in rolling a distance of 176 m.

111. Find the difference of the areas of two segments of a circle formed by a chord of length 5 cm subtending an angle of 90° at the centre.

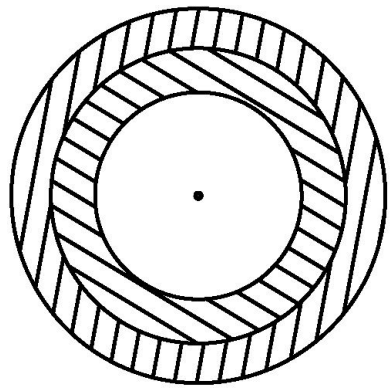
112. Find the difference of the areas of a sector of angle 120° and its corresponding major sector of a circle of radius 21 cm.

113. The central angles of two sectors of circles of radii 7 cm and 21 cm are respectively 120° and 40°. Find the areas of the two sectors as well as the lengths of the corresponding arcs.

114. The length of the minute hand of a clock is 5 cm. Find the area swept by the minute hand during the time period 6:05 a m and 6:40 a m.

115. All the vertices of a rhombus lie on a circle. Find the area of the rhombus, if area of the circle is 1256 cm^2. (Use $\pi = 3.14$).

116. An archery target has three regions formed by three concentric circles as shown in the below figure. If the diameters of the concentric circles are in the ratio 1: 2:3, then find the ratio of the areas of three regions.



117. Area of a sector of central angle 200° of a circle is 770 cm^2. Find the length of the corresponding arc of this sector.

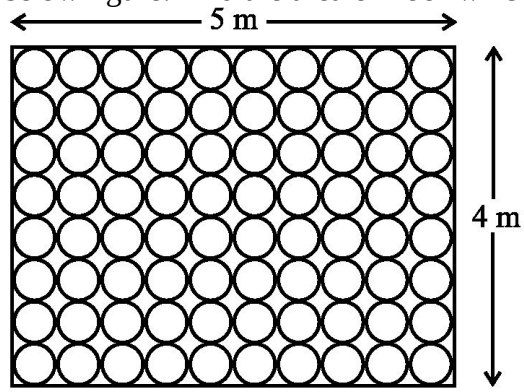
118. Three circles each of radius 3.5 cm are drawn in such a way that each of them touches the other two. Find the area enclosed between these circles.

119. Find the area of the sector of a circle of radius 5 cm, if the corresponding arc length is 3.5 cm.

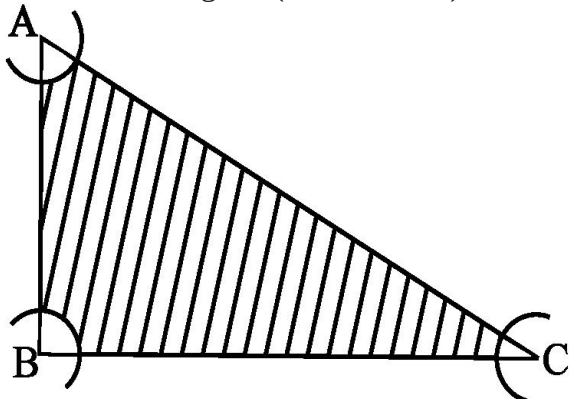
120. Four circular cardboard pieces of radii 7 cm are placed on a paper in such a way that each piece touches other two pieces. Find the area of the portion enclosed between these pieces.

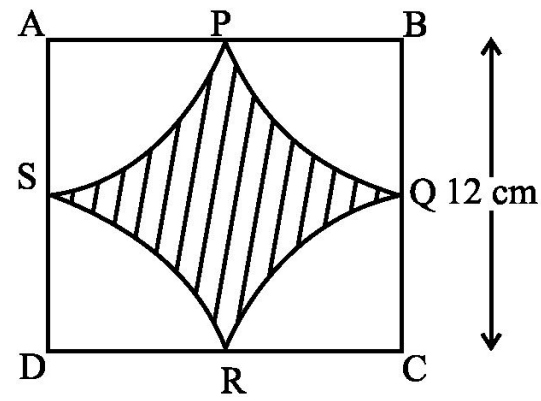
121. On a square cardboard sheet of area 784 cm^2, four congruent circular plates of maximum size are placed such that each circular plate touches the other two plates and each side of the square sheet is tangent to two circular plates. Find the area of the square sheet not covered by the circular plates.

122. Floor of a room is of dimensions 5 m × 4 m and it is covered with circular tiles of diameters 50 cm each as shown in the below figure. Find the area of floor which is not covered by tiles.



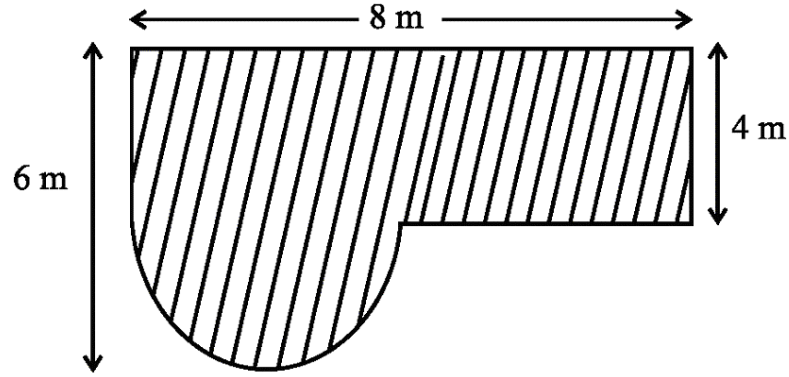
123. With the vertices A, B and C of a triangle ABC as centres, arcs are drawn with radii 5 cm each as shown in below figure. If AB = 14 cm, BC = 48 cm and CA = 50 cm, then find the area of the shaded region. (Use $\pi = 3.14$).



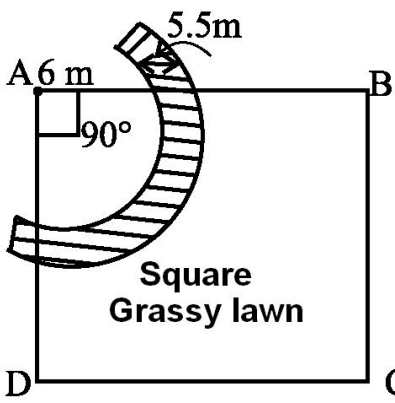


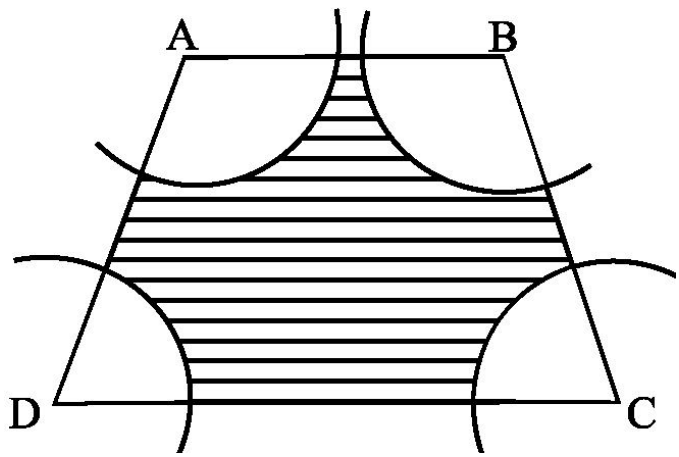
124. Find the area of the shaded region in the above right sided figure, where arcs drawn with centres A, B, C and D intersect in pairs at mid-points P, Q, R and S of the sides AB, BC, CD and DA, respectively of a square ABCD (Use $\pi = 3.14$).

125. Find the area of the shaded field shown in the below figure.



126. A calf is tied with a rope of length 6 m at the corner of a square grassy lawn of side 20 m. If the length of the rope is increased by 5.5m, find the increase in area of the grassy lawn in which the calf can graze.



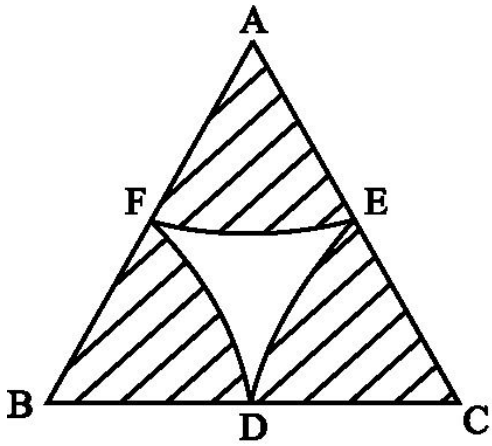


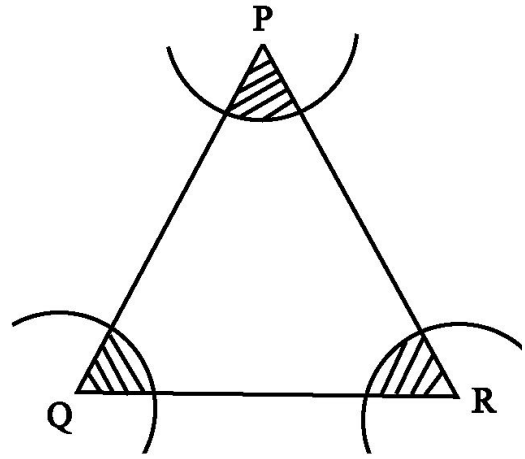
127. In the above right sided figure, ABCD is a trapezium with AB || DC, AB = 18 cm, DC = 32 cm and distance between AB and DC = 14 cm. If arcs of equal radii 7 cm with centres A, B, C and D have been drawn, then find the area of the shaded region of the figure.

128. A circular pond is 17.5 m is of diameter. It is surrounded by a 2 m wide path. Find the cost of constructing the path at the rate of Rs 25 per m^2

129. A circular park is surrounded by a road 21 m wide. If the radius of the park is 105 m, find the area of the road.

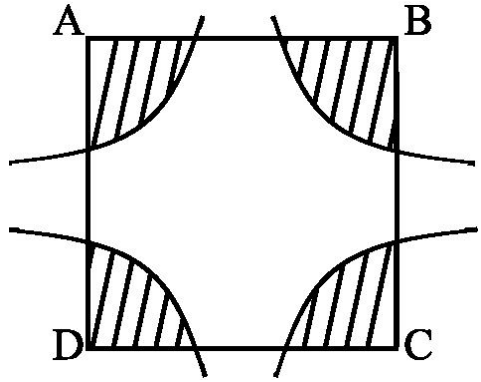
130. In the below figure, arcs are drawn by taking vertices A, B and C of an equilateral triangle of side 10 cm. to intersect the sides BC, CA and AB at their respective mid-points D, E and F. Find the area of the shaded region (Use $\pi = 3.14$).





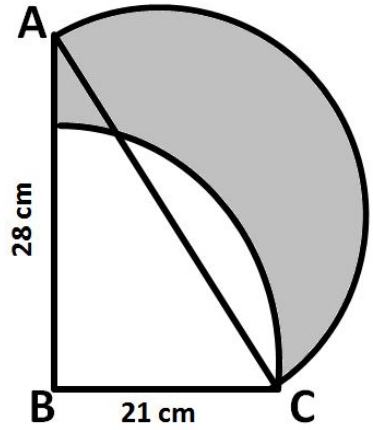
131. In the above right sided figure, arcs have been drawn with radii 14 cm each and with centres P, Q and R. Find the area of the shaded region.

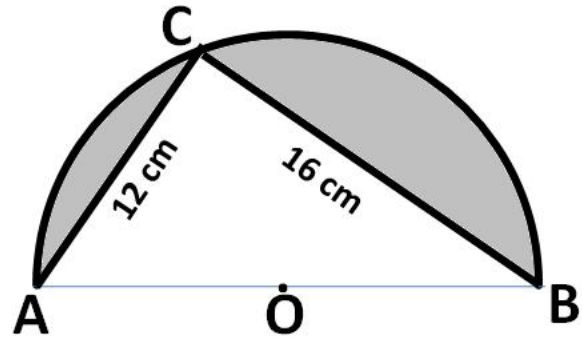
132. In the below figure, arcs have been drawn of radius 21 cm each with vertices A, B, C and D of quadrilateral ABCD as centres. Find the area of the shaded region.



133. A piece of wire 20 cm long is bent into the form of an arc of a circle subtending an angle of 60° at its centre. Find the radius of the circle.

134. In the below figure, ABC is a right angled triangle at B, AB = 28 cm and BC = 21 cm. With diameter a semicircle is drawn and with BC as radius a quarter circle is drawn. Find the area of the shaded region correct to two decimal places.





135. In the above right-sided figure, O is the centre of a circular arc and AOB is a straight line. Find the perimeter and the area of the shaded region. (use $\pi = 3.142$)